

Lab Assignment-3

Computer Architecture (CS F342)

Semester-I, 2026-27

Department of Computer Science and Information Systems (CSIS)

BITS-Pilani, K K Birla Goa Campus, Goa, India.

Due time: Sept 10, 2025: :13:00

Marks:20

/*#####

Please don't upload the assignments, template file/solution and lab. manual on GitHub or others public repository. Kindly remove them, if you have uploaded the previous assignments. It violates the BITS's Intellectual Property Rights (IPR).

*****/

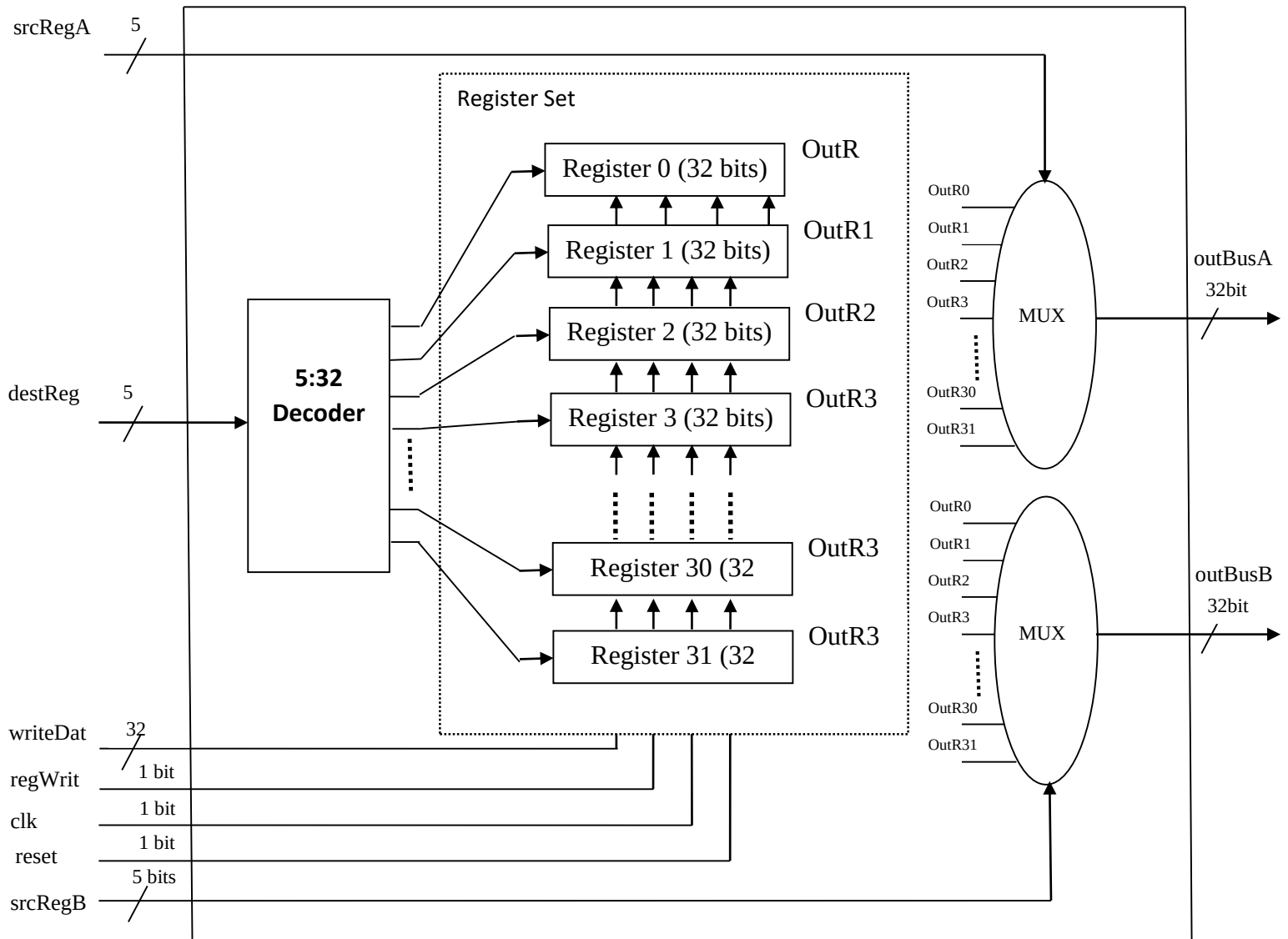
Download the *CA_Lab-3_Template.zip* and it contains 3 problems (P1, P2 and P3 folders) and rename it as *<CampusID>_ CA_Lab-3*.

One's lab marks depend on one's performance on lab viva even if one has no complete working code. Serious action will be taken if one has the fully working code but is unable to explain it during the viva.

P1) Design and implement the following module using verilog HDL with the following specifications.

Marks Distribution:

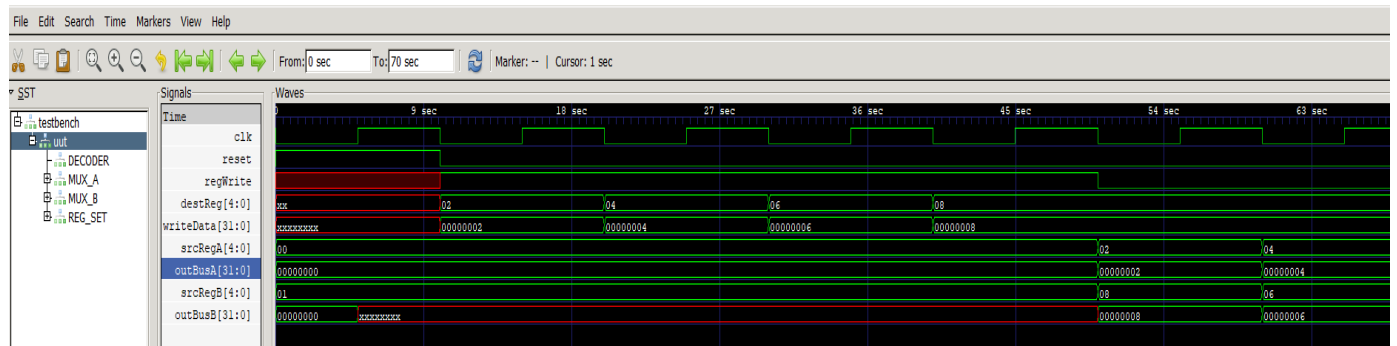
File name	Module	Marks
dff.v	dff	0
register32bit.v	register32bit	0.5
registerSet32.v	registerSet	0.5
decoder5to32bit.v	decoder_5to32	1.5
mux32to1_1bit.v	Mux32to1_1bit	1
mux32to1_32bit.v	Mux32to1_32bit	0.5
topModule_registerFile.v	registerFile	0.5
testbench.v	testbench	0.5



Total marks will be awarded to your design based on Viva. If you cannot answer the question(s) asked by the evaluator, then the marks will be deducted accordingly by the evaluator, which is the final decision.

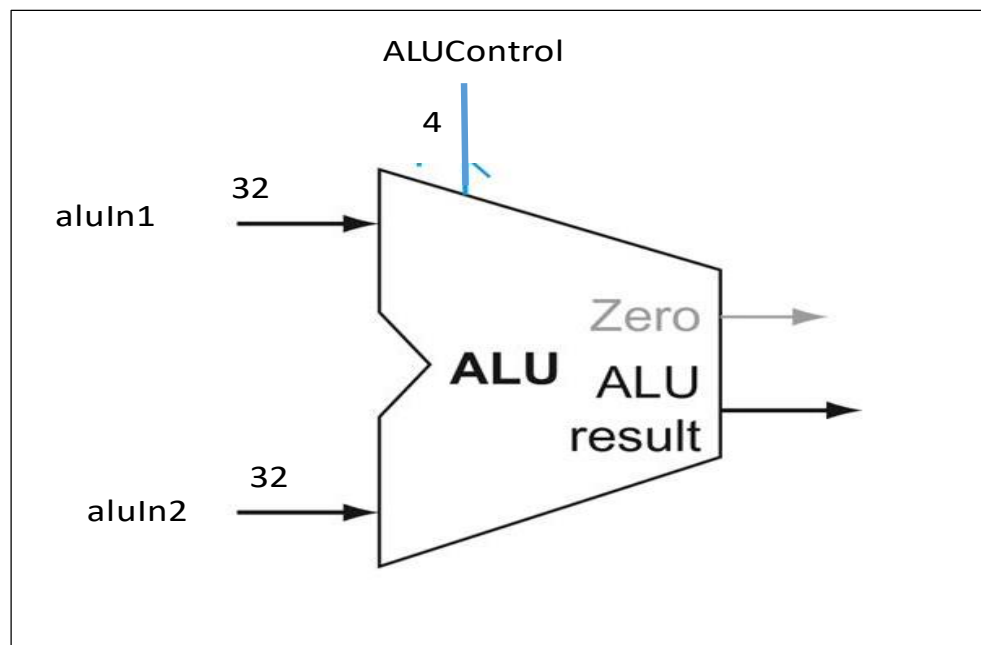
Write the **structural** (not the behavioral) Verilog HDL code for the above modules. The register 0 holds the 0 value always.

Expected Output:



Upload all files including output in the folder.

P2) Design MIPS's ALU using behavioural Verilog HDL code. It will perform all the operation mention in the below.



Instruction opcode	ALUOp	Instruction operation	Funcnt field	Desired ALU action	ALU control input
LW	00	load word	XXXXXX	add	0010
SW	00	store word	XXXXXX	add	0010
Branch equal	01	branch equal	XXXXXX	subtract	0110
R-type	10	add	100000	add	0010
R-type	10	subtract	100010	subtract	0110
R-type	10	AND	100100	AND	0000
R-type	10	OR	100101	OR	0001
R-type	10	set on less than	101010	set on less than	0111

ALUOp		Funct field						Operation
ALUOp1	ALUOp0	F5	F4	F3	F2	F1	F0	
0	0	X	X	X	X	X	X	0010
X	1	X	X	X	X	X	X	0110
1	X	X	X	0	0	0	0	0010
1	X	X	X	0	0	1	0	0110
1	X	X	X	0	1	0	0	0000
1	X	X	X	0	1	0	1	0001
1	X	X	X	1	0	1	0	0111

Marks Distribution:

Module	Marks
ALU	3
aluCtrl	3
testbench	3

Upload all files including output in the folder.

P3) Write an ASM program using MIPS ISA to check whether a number, N (take it from the user), is Fibonacci. Print the proper message for this, and if yes, find and print its position in the sequence. **Do not define any function.** Use the MARS simulator.

Marks: 5 + 1

Upload all files including output in the folder.

Instructions for uploading the results:

Make a zip file, <CampusID>_CA_Lab-3.zip which can contain the *all files*.

Do not create archives in other formats (rar, tar.gz etc). Once uploaded on Quanta, remember to **submit** for grading. Do not leave it as a draft.

/*#####

Note: Please don't upload the assignments, template file/solution and lab. manual on GitHub or others public repository.

It violates the BITS's Intellectual Property Rights (IPR).

*****/